

Course Code:	CSC 315
Course Title:	Object Oriented Programming - Lab
Credit Hours:	4 (3-1)
Program(s) & Group(s):	BSDS, BSSE, BSAI, BCS, BS Cybersecurity
Academic Calendar Semester:	2 nd Semester
Lecture Timing:	As per schedule
Prerequisites and/or Expectations:	Programming Fundamentals
Instructor:	Mr. Ali Haider
Website (if any):	https://imsciences.edu.pk/faculty/?act=faculty_profile_view&teacher_id1=211
Email:	ali.haider@imsciences.edu.pk
Office Location:	Opposite to PHD Room – Basement Floor
Office Contact Hours:	8:00 pm- 4:00 pm (Monday-Friday)
Course Description:	The main purpose of this lab is to introduce students to object-oriented programming (OOP) skills. It provides detailed knowledge of OOP concepts and techniques using C++. The lab covers important topics such as classes, function overloading, data abstraction, information hiding, encapsulation, inheritance, polymorphism, file handling and exception handling.

Course Objectives (COs):	At the end of the course, the students will be able:			
	1. To enable students to construct, execute, and test basic programming solutions using core Object-Oriented Programming concepts within an integrated development environment (IDE). 2. To provide students with hands-on experience in implementing and debugging intermediate OOP concepts, including inheritance, polymorphism, and exception handling. 3. To enable students to integrate and develop complete, well-structured console applications by combining core and intermediate OOP features.			
Course Learning Outcomes (CLOs):	At the end of the course the students will be able to:		Domain Level	BT Domain
	CLO1:	Demonstrate the proficiency to write, compile, and execute basic programs using classes, objects, encapsulation, and constructors within an Integrated Development Environment (IDE).	P3	Precision
	CLO2:	Construct programs that implement inheritance, composition, and polymorphism to solve structured computing problems.	P3	Manipulation/ Precision
	CLO3:	Integrate core and intermediate OOP concepts to build complete, functional console applications that include exception handling and file handling.	P4	Articulation
	* BT= Bloom's Taxonomy, C=Cognitive domain, P=Psychomotor domain, A= Affective domain			
Alignment of CLOs and COs:		CO1	CO2	CO3
	CLO-1	✓		
	CLO-2		✓	
	CLO-3			✓
Course Assessment(s):	As per IMSciences policy • Mid-term Exam (30%) • Assignment (20%) • Final-term Exam (50%)			
	Sno.	CLOs	Relevant Assessment Methods	
	1	CLO 1	Mid Term	
	2	CLO 2	Final Term	
	3	CLO 3	Final Term	

Graduate Attributes (GAs)/ Program Learning Outcomes (PLOs):	1.	Gain an understanding of the underpinning theories of fundamental principles and technologies within the area of computer science (Academic education).
	2.	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements (Knowledge for Solving Computing Problems).
	3. Lab	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and relevant domain disciplines (Problem Analysis).
	4. Lab	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations (Design/ Development of Solutions).
	5. Lab	Create, select, adapt, and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations (Modern Tool Usage).
	6.	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings (Individual and Teamwork).
	7.	Communicate effectively with the computing community and with society about complex computing activities by being able to comprehend and write effective reports, instructions (Communication).
	8.	Understand and assess societal, health, safety, legal, and cultural issues within local and global contexts, and the consequential responsibilities relevant to professional computing practice (Computing Professionalism and Society).

	9.	Understand and commit to professional ethics, responsibilities, and norms of professional computing practice (Ethics).									
	10.	Recognize the need, and have the ability, to engage in independent learning for continual development as a computing professional (Life-long Learning).									
Alignment of CLOs and PLOs/GAs:		GA1	GA2	GA3	GA4	GA5	GA6	GA7	GA8	GA9	GA10
	CLO-1					✓					
	CLO-2			✓							
	CLO-3				✓						
Course Resources:	<i>The background texts on which the course is based are:</i> Text Books: 1. Lab Manual										
Course Teaching Methodology:	The course will be taught through lab demonstrations, hands-on practice tasks during class, and guided lab tasks to build skills step by step. Students will also complete home practice tasks to strengthen their understanding and improve their programming abilities.										
Course Assignment/Quizzes /Presentation:	Students will be evaluated through a mid-term exam, a final-term exam, and/or a semester project. Each student must also give a presentation to defend their project and show their understanding of the work, whether it is done individually or in a group.										
Attendance Policy	- As per the IMSciences policy students will not be allowed to sit in the final-Term exam if they fail to attend less than 75% of the total classes. Attendance will be marked after 15 minutes of the class starts. - If a student comes into the class after 15 minutes, he/she may sit in the class, but their attendance will not be marked for that class. - In case of emergencies or unavoidable circumstances students have to report me directly or e-mail.										
General Policies and Behavioral Expectations:	<i>During the class sessions students are expected to:</i> Ensure timely arrival in class and remain present throughout the class session.										

	▪ Participate in class discussions and engage in any individual or group tasks. ▪ Adhere to institute attendance policy and general code of conduct. ▪ Maintain and update record of all class notes, handouts, and relevant materials. ▪ Adhere to basic principles of academic integrity with regards to exams and assignments ▪ Use the lab equipment properly and carefully. ▪ Students are not allowed to play games or use any offline/online application or other media (news, social media, movies etc.) in computer labs/classroom. ▪ Students must follow all the cyber laws and maintain other’s privacy while working in computer labs.		
Grades to be Assigned			
Grades	Marks (%)	Grade Points	Remarks
A+	91-100	4.0	Outstanding
A	87-90	4.0	Excellent
B+	80-86	3.5	Very Good
B	72-79	3.0	Good
C+	66-71	2.5	Satisfactory
C	60-65	2.0	Pass
F	Below 60	0	Fail

Week Wise Course Distribution

Week	Topic Description	CLO(s)
1.	Working with Structures	CLO 1
2.	Introduction to OOP Concepts: Classes & Objects	CLO 1
3.	Data Hiding and Encapsulation	CLO 1
4.	Basic Abstraction Implementation	CLO 1
5.	Constructors and Destructors	CLO 1
6.	Setters, Getters, Helpers & Display Functions	CLO 1
7.	Working with Static Members, Constant Functions & “this” pointer.	CLO 1
8.	Basic Inheritance	CLO 1
Mid-Term Examination		
9.	Multi-Level Inheritance & Multiple Inheritance	CLO 2
10.	Composition	CLO 2
11.	Polymorphism – Function Overloading	CLO 2
12.	Polymorphism – Function Overriding	CLO 2
13.	Abstract Classes (Virtual & Pure Virtual Functions)	CLO 2
14.	Advanced OOP Concepts: Friend Functions and Classes	CLO 2
15.	Integrating Exception Handling	CLO 3
16.	Integrating File Handling	CLO 3
Final Term Examination		